Exercise 1

(i) $f(x) = x-2$, $g(x) = \frac{1}{x}$

(ii) $f(x) = \sin(2x)$, $g(x) = x^4$

(iii) $f(x) = \sin(x)$, $g(x) = 2x^3 + 5x$

(iv) $f(x) = \cos(x)$, $g(x) = \frac{1}{2+x}$

Exercise 2

(i) $|\cos(x)| = \begin{cases} \cos(x), & 0 \le x < \frac{\pi}{2} \\ -\cos(x), & \frac{\pi}{2} \le x < \frac{3\pi}{2} \\ \cos(x), & \frac{3\pi}{2} \le x \le 2\pi \end{cases}$

(ii) $|x^2-4|+4 = \begin{cases} x^2, & x \le -2 \text{ or } x \geq 2 \\ 8-x^2, & -2 < x < 2 \end{cases}$

(iii) $|x-1| + |x| + |x+1| = \begin{cases} 3x, & x \geq 1 \\ 2x+2, & 0 \le x < 1 \\ x+2, & -1 \le x < 0 \\ -3x, & x < -1 \end{cases}$

Exercise 3

(i) $f \circ g = \sqrt{\sqrt{3-x} - 2}$, $D(f \circ g) = (-\infty, 1]$ and $R(f \circ g) = [0, \infty)$

$g \circ f = \sqrt{3 - \sqrt{x-2}}$, $D(g \circ f) = [2, 11]$ and $R(g \circ f) = [0, \sqrt{3}]$

(ii) $f \circ g = (\sqrt{x-3})^2 + 3 = x$, $D(f \circ g) = [0, \infty)$ and $R(f \circ g) = [0, \infty)$

$g \circ f = \sqrt{x^2 + \cancel{3} - \cancel{3}} = |x|$, $D(f \circ g) = (-\infty, \infty)$ and $R(f \circ g) = [0, \infty)$

(iii) $f \circ g = \frac{\ln^2(x)}{1 + \ln^2(x)}$, $D(f \circ g) = (-\infty, \infty)$ and $R(f \circ g) = [0, \infty)$

$g \circ f = \ln\left[\frac{x^2}{1+x^2}\right]$, $D(f \circ g) = (-\infty, \infty)$ and $R(f \circ g) = (-\infty, \infty)$

Exercise 4

(i) Let $f(x) = \frac{1}{x+1}$

$E(f(x)) = \frac{1}{2}\left[f(x) + f(-x)\right] = \frac{1}{2}\left[\frac{1}{x+1} + \frac{1}{(-x)+1}\right]$

$= \frac{1}{\cancel{2}}\left[\frac{\cancel{2}}{(1+x)(1-x)}\right] = \frac{1}{1-x^2}$

$O(f(x)) = \frac{1}{2}\left[f(x) - f(-x)\right] = \frac{1}{2}\left[\frac{1}{x+1} - \frac{1}{(-x)+1}\right] = -\frac{1}{\cancel{2}}\left[\frac{\cancel{2}x}{1-x^2}\right]$

$= \frac{-x}{1-x^2}$

$\therefore f(x) = \boxed{\frac{1}{1-x^2} - \frac{x}{1-x^2}}$

(ii) Let $f(x) = \sqrt{\left(\frac{1-x}{1+x}\right)}$

$E(f(x)) = \frac{1}{2}\left[\sqrt{\left(\frac{1-x}{1+x}\right)} + \sqrt{\left(\frac{1+x}{1-x}\right)}\right]$

$O(f(x)) = \frac{1}{2}\left[\sqrt{\left(\frac{1-x}{1+x}\right)} - \sqrt{\left(\frac{1+x}{1-x}\right)}\right]$

(iii) Let $f(x) = \sin(x+1)$

$E(f(x)) = \frac{1}{2}\left[\sin(x+1) + \sin(1-x)\right] = \frac{1}{2}\left[\sin(x)\cos(1) + \cancel{\cos(x)\sin(1)} + \sin(x)\cos(1) - \cancel{\sin(1)\cos(x)}\right]$

$= \sin(x)\cos(1)$

$O(f(x)) = \frac{1}{2}\left[\sin(x+1) - \sin(1-x)\right] = \frac{1}{2}\left[\sin(x)\cos(1) + \cancel{\cos(x)\sin(1)} - \cancel{\sin(1)\cos(x)} + \cos(1)\sin(x)\right]$

$= \sin(x)\cos(1)$

Exercise 5

(i) Let $f, g = x^n, x^m$: $f \circ g = (x^m)^n = x^{mn}$ and $g \circ f = (x^n)^m = x^{mn}$

Therefore $f \circ g \equiv g \circ f$

(ii) if f is an odd function, then $f(x) = \frac{1}{2}\left[f(x) - f(-x)\right]$

$\Rightarrow f(0) = \frac{1}{2}\left[f(0) - f(0)\right] = \frac{1}{2}(0) = 0$

Therefore, $f(0) = 0$

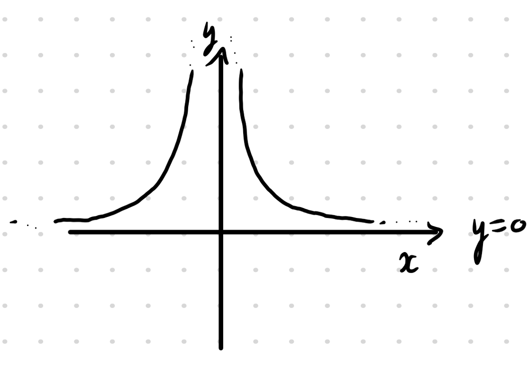
(iii) if f is both even and odd, then $f(x) = \frac{1}{2}\left[f(x) + f(-x)\right] = \frac{1}{2}\left[f(x) - f(-x)\right]$

$\Rightarrow 2f(-x) = 0$

Therefore $f(-x) = 0$

Exercise 6

$\frac{|x|}{x^2} = \frac{\sqrt{x^2}}{x^2} = \sqrt{\frac{1}{x^2}} \Rightarrow D(k): x \in R, x \neq 0$



$y=0$ asymptote

Exercise 7

f is defined $\forall x \in R$, g is defined $\forall x \in R^+$ or $x \geq 0$

(i) $f + g = x^3 + x + x^{\frac{1}{4}}$, defined for $x \geq 0$

(ii) $g \circ g = x^{\frac{1}{16}}$, defined for $x \geq 0$

(iii) $g \circ f = (x^3 + x)^{\frac{1}{4}}$, defined $\forall x \in R, x \neq 0$

(iv) $g \circ \frac{1}{f} = \left(\frac{1}{x^3+x}\right)^{\frac{1}{4}}$, defined for $\forall x \in R, x \neq 0$

Exercise 8

(i) $f(x) = x^2$ and $g(x) = \cos(x)$

(ii) $f(x) = x$ and $g(x) = \sin(x)$

(iii) for a given function $f(x)$, f is periodic if $f(x) \equiv f(x+k)$ $\forall k \in R$

(iv) Let f be a function

if $x \le y$ and $f(x) \le f(y)$, the function is monotonic increasing $(f(x) = x)$

if $x \le y$ and $f(x) \geq f(y)$, the function is monotonic decreasing $(f(x) = -x)$

Exercise 9

(i) $2\cos^2(x) + 3\sin(x) = 3$

$\Rightarrow 2\left[1 - \sin^2(x)\right] = 3 - 3\sin(x)$

$\Rightarrow 2\sin^2(x) - 3\sin(x) + 1 = 0$

Let $u = \sin(x)$: $2u^2 - 3u + 1 = 0$

Thus, $u = \frac{3 \pm \sqrt{(3)^2 - 4(2)(1)}}{2(2)} = \frac{3}{4} \pm \frac{1}{4}$

$\therefore \sin(x) = 1$ and $\sin(x) = \frac{1}{2}$

$\therefore x = 0$ and $x = \frac{\pi}{6}$

(ii) $\cosh^2(t) + \sinh^2(t) = \left(\frac{e^t + e^{-t}}{2}\right)^2 + \left(\frac{e^t - e^{-t}}{2}\right)^2$

$= \frac{1}{4}\left[e^{2t} \cancel{+2} + e^{-2t} + e^{2t} \cancel{-2} + e^{-2t}\right]$

$= \frac{1}{2}\left(e^{2t} + e^{-2t}\right) = \cosh(2t)$

$\therefore \sinh^2(t) + \cosh^2(t) = \cosh(2t)$

(iii) $3\sinh(x) - \cosh(x) = 1 \Rightarrow 3\left(\frac{e^x - e^{-x}}{2}\right) - \left(\frac{e^x + e^{-x}}{2}\right) = 1$ (Hyperbolic definitions)

$\Rightarrow 4e^x - 1 - 2e^{-x} = 0$

$\Rightarrow 4e^{2x} - e^x - 2 = 0$

Let $u = e^{2x}$: $4u^2 - u - 2 = 0$

Thus, $u = \frac{1 \pm \sqrt{(-1)^2 - 4(4)(-2)}}{2(4)} = \frac{1}{8} \pm \frac{\sqrt{33}}{8}$

Therefore, $x = \ln\left(\frac{1 \pm \sqrt{33}}{8}\right)$

(iv) RTP $\forall x \in R$, $\sinh^{-1}(x) = \ln(x + \sqrt{x^2+1})$

Let $y = \sinh(x) = \frac{e^x - e^{-x}}{2}$ (Hyperbolic definition)

$\Rightarrow 2y = e^x - e^{-x}$

$\Rightarrow e^{2x} - 2ye^x - 1 = 0$

Thus, $e^x = \frac{2y \pm \sqrt{(2y)^2 - 4(1)(-1)}}{2(1)} = y \pm \sqrt{y^2+1}$

$\Rightarrow x = \ln\left[y + \sqrt{y^2+1}\right]$

Therefore $\sinh^{-1}(x) = \ln\left[x + \sqrt{x^2+1}\right]$

Exercise 10

(i) Referencing

Since g is monotonic decreasing, a greater input leads to a smaller output.

Now because the output increases with the input for f which is monotonic increasing, it implies that the opposite is also true where the output decreases as the input decreases.

$f \circ g$ must also be monotonic decreasing as the input for f is always decreasing and the direction for f follows from the input.

$f \circ f \circ g$ follows through with the same logic.

Therefore, you can consider the monotonic logic to be similar to that of positive and negative numbers where the negative changes the sign, whilst the positive doesn't.

Consider $y \in D(g)$: $g(x) > g(y)$ with $x < y$ (definition of monotonic decreasing function)

Since f: $f(x) < f(y)$ with $x < y$, then it also follows that $f(x) > f(y)$ with $x > y$ (definition of monotonic increasing function)

Thus, $f(x-k) < f(x)$ $\forall k \in R^+$, which follows through for $f \circ g$ logically.

Therefore, $f \circ g$ is monotonic decreasing.